

Lecture 3: Introducing Uncertainty and Time

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3.1 Modelling Uncertainty as Lotteries

Definition 3.1 (Simple lottery) A simple lottery over outcomes $X = \{x_1, x_2, \dots, x_n\}$ is defined as a probability distribution $p = (p(x_1), p(x_2), \dots, p(x_n))$, where $p(x_k) \geq 0$ is the probability that x_k occurs and $\sum_{k=1}^n p(x_k) = 1$.

Example 3.2 Consider a division manager who has to decide on whether a research and development (R&D) project is worthwhile. The manager has two options – g for go ahead with R&D or s for status quo – so $A = \{g, s\}$. Imagine that there are only two final outcomes: his product line is successful, which is equivalent to a profit of 10 (choose your denomination), or his product line is obsolete, which is equivalent to a profit of 0. Lastly, there is a real cost associated with the R&D campaign, so if the manager chooses g , the payoff will be deducted by 1 so that $X = \{-1, 0, 9, 10\}$. If we assume that $u(x) = x$ such that the payoff function

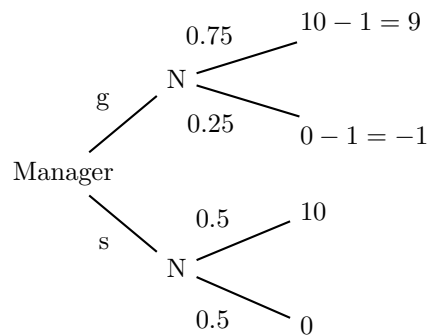


Figure 3.1: The R&D problem with costs.

10 means \$10. This means if the manager chooses g , $\mathbb{E}(u(x)) = \mathbb{E}(x) = 0.75(9) + 0.25(-1) = 6.5$. Notice we might also use the notation $\mathbb{E}(u(x)|g)$ to indicate that given the manager has chosen g , the expected value is 6.5. Similarly, $\mathbb{E}(u(x)|s) = 0.5(10) + 0.5(0) = 5$, so a rational player would choose the action with the highest expected payoff.

Now we will explore if $u(x) = x$ where x is the amount of money is reasonable.

3.1.1 Risk Attitudes

So far we have only considered the expected payoff as the deciding factor on which choice to make; as rational people, we choose the option with the highest expected payoff. However, what if the two options have the same expected payoff, what do we do then.

Example 3.3 *Imagine a lottery with $x_1 = \$4$, $x_2 = \$9$, and $x_3 = \$16$, with associated payoff values p_1, p_2, p_3 . Then the expected value of this lottery is*

$$\mathbb{E}[u(x)|p] = \sum_{k=1}^3 p_k u(x_k) = p_1 u(x_1) + p_2 u(x_2) + p_3 u(x_3).$$

Now we let $p' = (p'_1, p'_2, p'_3) = (\frac{7}{12}, 0, \frac{5}{12})$ and $\hat{p} = (\hat{p}_1, \hat{p}_2, \hat{p}_3) = (0, 1, 0)$. This means that lottery p' randomizes between x_1 and x_3 ; whereas, lottery $\hat{p}$ is a sure bet that you will win \$9. To decide which lottery to pick, we must compute the expected value and pick the larger one. if $u(x) = x$,

$$\frac{7}{12}u(x_1) + \frac{5}{12}u(x_3) = \frac{28}{12} + \frac{80}{12} = 9; \quad u(x_2) = 9.$$

This means that the average payoffs of the two lotteries are the same. If someone chooses p' they run the risk of getting \$5 less, but also have the chance of getting \$7 more!

Definition 3.4 (Risk Attitudes) *If the player is indifferent between the two lotteries, meaning $\frac{7}{12}u(4) + \frac{5}{12}u(16) = u(9)$, that player is risk neutral. Alternatively, the player would be called risk averse if they may prefer not to be exposed to risk for the same expected payout, so $\frac{7}{12}u(4) + \frac{5}{12}u(16) < u(9)$. In other words, they would want to take the sure payout then be subjected to a nondegenerate lottery. Lastly, if the player would prefer to be exposed to a lottery with the same expected payoff, they are risk loving.*

Remark 3.5 *This was all assuming that $u(x) = x$, which isn't always a good assumption. If $u(x) = x$, then like above, the person is risk neutral: $\frac{7}{12}u(4) + \frac{5}{12}u(16) = \frac{7}{12}4 + \frac{5}{12}16 = u(9) = 9$. The preference relationship is $\$16 \succ \$9 \succ \$4$.*

However, now assume that $u(x) = \sqrt{x}$. The ordinal ranking is preserved, but $\frac{7}{12}u(4) + \frac{5}{12}u(16) = \frac{7}{12}2 + \frac{5}{12}4 = \frac{17}{6} < 3 = u(9)$. In summary, the player with the modified payoff function, which preserves the ordinal ranking among the outcomes, will exhibit different risk attitudes.

3.1.2 St. Petersburg Paradox

Example 3.6 (St. Petersburg Paradox) *Consider flipping a fair coin (i.e. $\mathbb{P}(T) = \mathbb{P}(H) = 0.5$) until a tails, T , appears. The payoff doubles with every occurrence of heads, and we start with a payoff of \$1, so if we flip T , we get \$1, if we flip HT , we get \$2, HHT we get \$4, etc.*

Now the decision problem: you get to choose between playing the game or \$1,000 immediately. What would you do? Discuss with your neighbor.

Remark 3.7 *If we immediately flip a T , then we end with \$1, and since $\mathbb{P}(T) = 0.5$, we would have the payoff for flipping a T is $0.5(1) = 0.5$. If we make it to the second round, we must have first flipped a H , which happens with $\mathbb{P}(H) = 0.5$. The probability we get out this round is still 50-50 since we get out if we*

roll a tail. For HT, we would have 0.25 probability of rolling that with payoff \$2. For HH, we go to the next round, which again has a 50-50 chance of ending. Putting this all together, we have

$$\mathbb{E}[u(x|\text{play game})] = \sum_{i=0}^{\infty} \underbrace{2^i}_{\text{payoff at the } i^{\text{th}} \text{ round}} \overbrace{\left(\frac{1}{2}\right)^i}^{\text{flipping } i \text{ H's in a row}} \overbrace{\left(\frac{1}{2}\right)}^{\text{flipping a T}} = \sum_{i=0}^{\infty} \frac{1}{2} = \infty.$$

This means that the expected value is infinite money. Does this change your mind? This all relies on the assumption that $u(x) = \$x$, which isn't always the case.

Example 3.8 We saw the St. Petersburg Paradox which says that by playing the game, we will make infinite money under the assumption that $u(x) = x$. However, we need to factor in that the more money we have, the less we desire it, an assumption we call the “decreasing marginal returns”. Also, this doesn't assume that by inflation, money loses its value over time. This means that we should look at functions that are increasing at a decreasing rate such as $u(x) = \sqrt{x}$ or $u(x) = \ln(x)$.

For the first case, we have $u(x) = \sqrt{x}$

$$\begin{aligned} \mathbb{E}[u(x)] &= \sum_{i=0}^{\infty} u(2^i) \left(\frac{1}{2}\right)^{i+1} \\ &= \sum_{i=0}^{\infty} \sqrt{2^i} \left(\frac{1}{2}\right)^{i+1} \\ &= \frac{1}{2} \sum_{i=0}^{\infty} \left(\frac{\sqrt{2}}{2}\right)^i \\ &= \frac{1}{2} \left(\frac{1}{1 - \frac{1}{\sqrt{2}}}\right) \\ &= \frac{\sqrt{2}}{2(\sqrt{2} - 1)} = 1 + \frac{1}{\sqrt{2}} \approx 1.70711 \end{aligned}$$

For the second case, we have $u(x) = \ln x$

$$\begin{aligned} \mathbb{E}[u(x)] &= \sum_{i=0}^{\infty} u(2^i) \left(\frac{1}{2}\right)^i \\ &= \sum_{i=0}^{\infty} \ln(2^i) \left(\frac{1}{2}\right)^{i+1} \\ &= \frac{1}{2} \sum_{i=0}^{\infty} i \ln(2) \left(\frac{1}{2}\right)^i \\ &= \frac{\ln(2)}{2} \sum_{i=0}^{\infty} i \left(\frac{1}{2}\right)^i \\ &= \frac{\ln(2)}{2} \frac{1}{1/2} \quad \text{by expectation of geometric series } p = 1/2 \\ &= \ln(2) \approx 0.69315 \end{aligned}$$

This makes sense that $u(x) = x^{1/2}$ has a higher expected payoff because it grows faster than $u(x) = \ln(x)$. But these two values are much much smaller than \$1,000, so those who said that they would rather have \$1,000 instead of playing the game have a keen sense of rationality.

3.2 Decisions over time

3.2.1 Backward induction

Once again, let's consider a modification of the R&D problem described in Figure 3.2, with a slight twist. Imagine that after the stage at which Nature first randomizes over whether or not the R&D project succeeds, the player faces another decision of whether to engage in a marketing campaign (m) or do nothing (d). The campaign can be launched only if the R&D project was executed. The marketing campaign will double the profits if the new product line is a success but not if it is a failure, and the campaign costs 6. The resulting modified decision problem is presented in Figure 3.2.1. For example, if the R&D effort succeeds and the marketing campaign is launched then profits from the product line will be 20 (double the original 10 in Figure 3.2), but the cost of the campaign, equal to 6, must be accounted for. This explains the payoff of $20 - 6 = 14$.

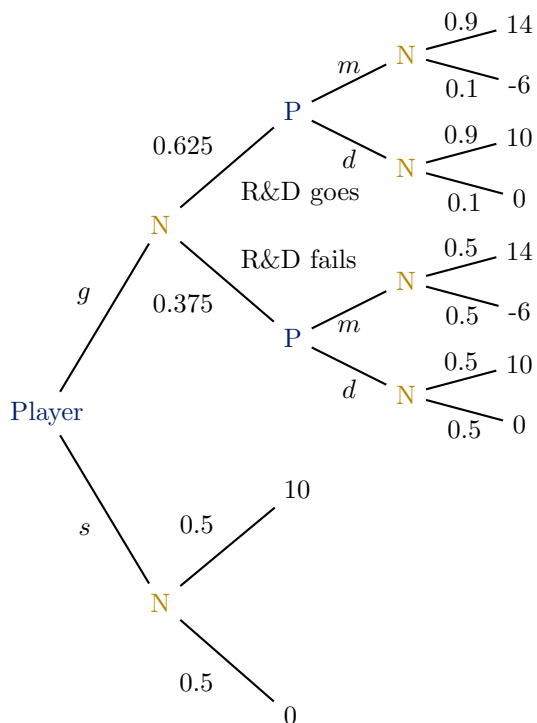


Figure 3.2: The R&D problem with a marketing phase.

Example 3.9 This is hard to decide what the Player will do as there are many stages of choices, and obviously Player has to decide between g and s before they have to decide about m or d . However, Player is rational, which means at each decision, they will make rational decisions. This means that we can start

from the back and work our way forward in a process called backward induction or dynamic programming. Since the full information of the game is known, we will start.

First, let's say that R&D succeeds. The player essentially has the choice between two lotteries— one materialized by choosing m and the other by choosing d . The Player is rational, so they will choose the lottery that gives them the largest expected payoff.

$$\mathbb{E}[u(x)|R\&D \text{ succeeds and } m] = 0.9(14) + 0.1(-6) = 12.6 - 0.6 = 12$$

$$\mathbb{E}[u(x)|R\&D \text{ succeeds and } d] = 0.9(10) + 0.1(0) = 9 - 0 = 9$$

which means that the Player will choose m with an expected payoff of 12, since $12 > 9$. Now we look at what happens if R&D fails, since the Player still gets to choose between m or d after figuring out how R&D went.

$$\mathbb{E}[u(x)|R\&D \text{ fails and } m] = 0.5(14) + 0.5(-6) = 7 - 3 = 4$$

$$\mathbb{E}[u(x)|R\&D \text{ fails and } d] = 0.5(10) + 0.5(0) = 5 - 0 = 5$$

which implies that the Player will choose d with the expected payoff of 5.

This means that we can rewrite the tree without those lotteries, as we figured out what the Player will do with what expected payoff. This reduced tree is seen in figure 3.9.

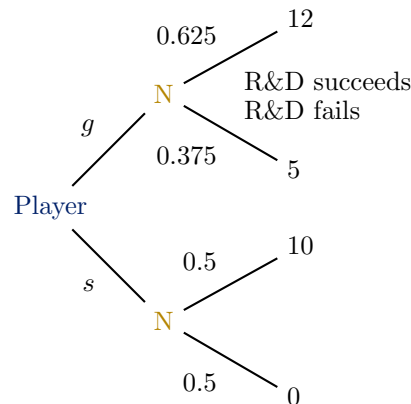


Figure 3.3: The R&D problem with a marketing phase.

Now we see that the player again has a choice between two lotteries, materialized by their choice of g or s . The rational player will choose the option which maximizes their expected payoff. This is exactly what we have analyzed before. namely $v(s) = 0.5(10) + 0 = 5$, and $v(g) = 0.625(12) + 0.375(5) = 9.375$, which is greater than 5, so the Player will choose g , wait to see the option of the R&D project and then choose accordingly.